

HVAC System Optimization Report

Ottawa Commercial Building Case Study

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HVAC Optimization Project

Realistic TMYx Weather • Part-Load Curves • Monte Carlo Analysis

Executive Summary

Baseline Performance

Annual Energy Consumption : 636,413 kWh
Annual Operating Cost : \$76,369.55

Best Scenario – Combined Measures

Annual Energy : 520,613 kWh
Energy Reduction : 115,800 kWh (18.2%)
Cost Savings : \$13,896.02
Payback Period : 3.6 years
CO₂ Reduction : 57.9 tonnes

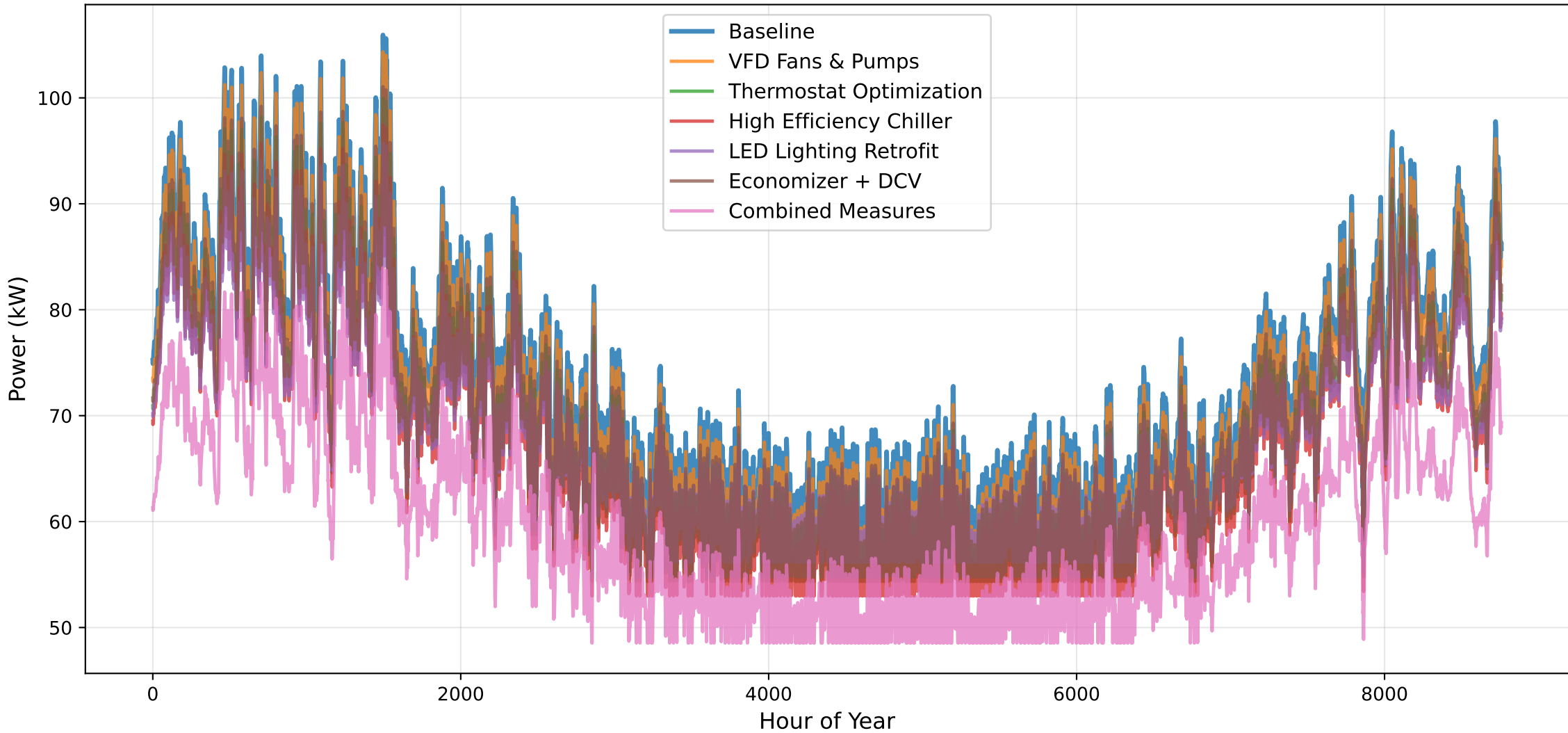
Parametric Optimization (SciPy)

Additional Annual Savings : \$-8,969
Minimum Annual Cost Found : \$53,504.14

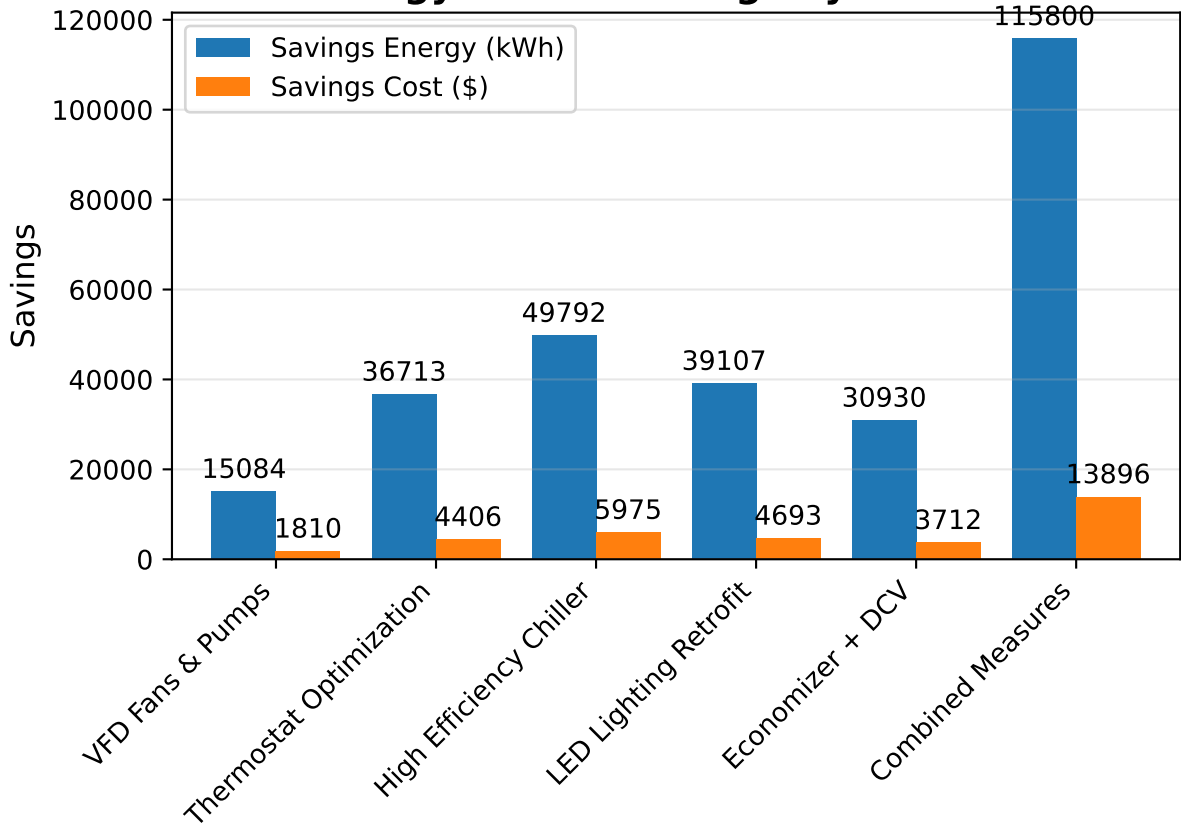
Monte Carlo Analysis (5,000 runs)

Mean ROI : 3.8 years
95% Confidence Interval : 2.0 – 6.8 years
Probability of ROI < 8 yrs : 99.0%

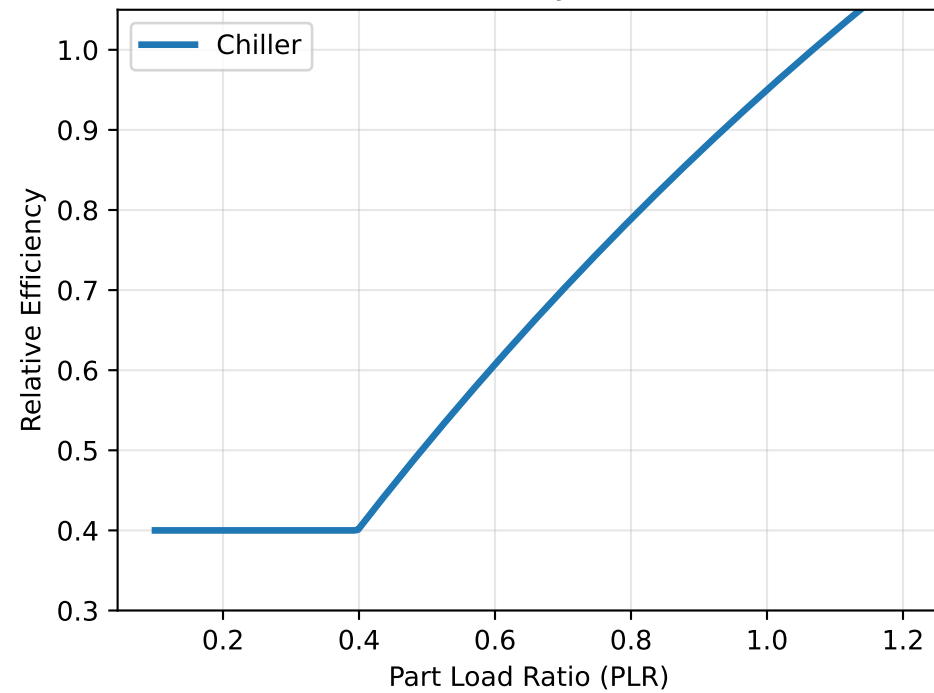
Hourly HVAC Energy Consumption Over One Year



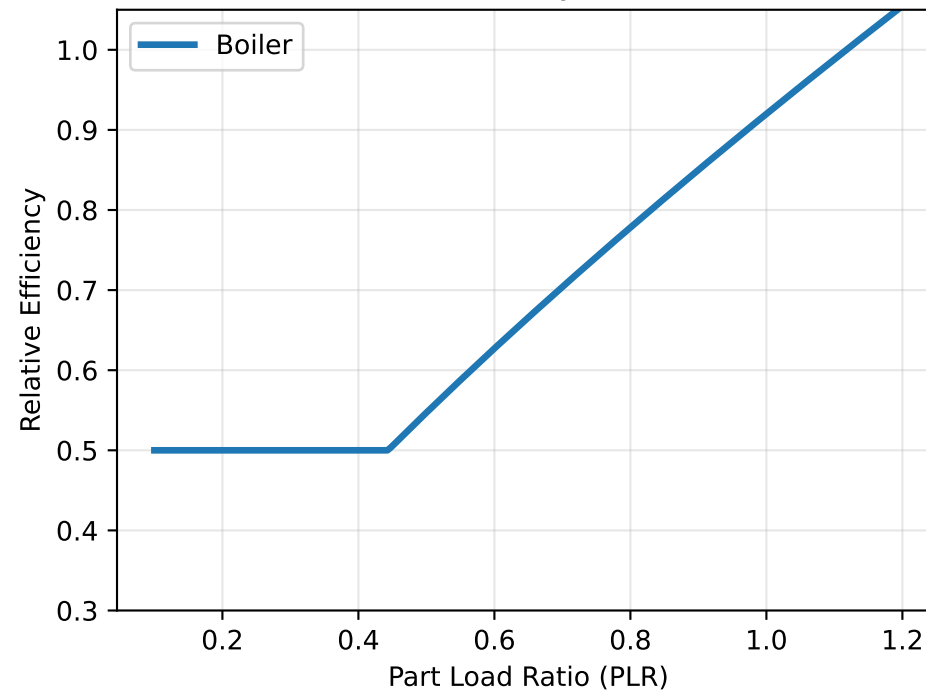
Energy & Cost Savings by Scenario



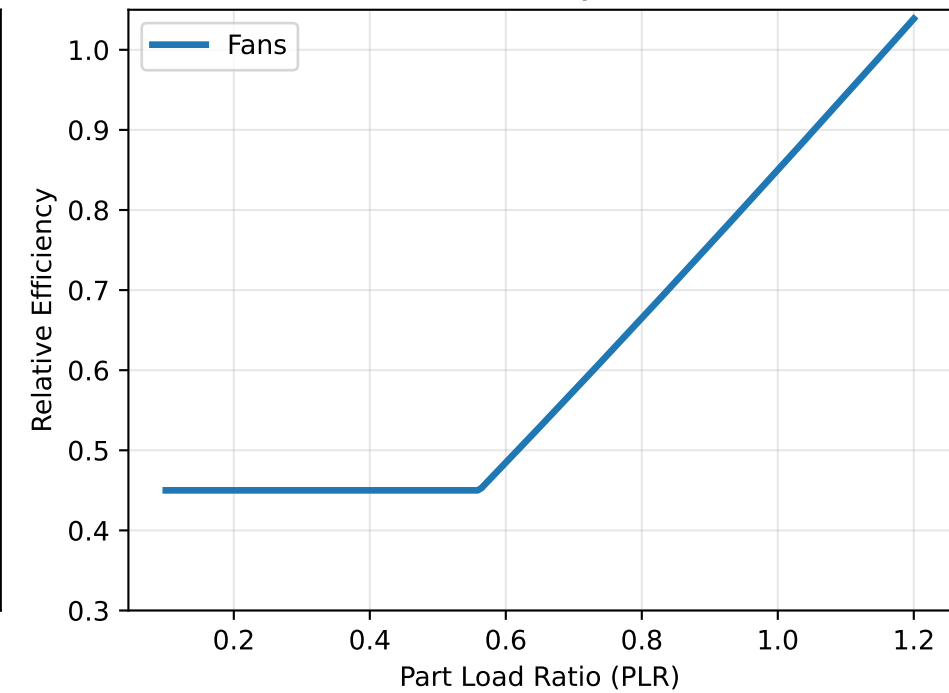
Part-Load Efficiency Curve: Chiller



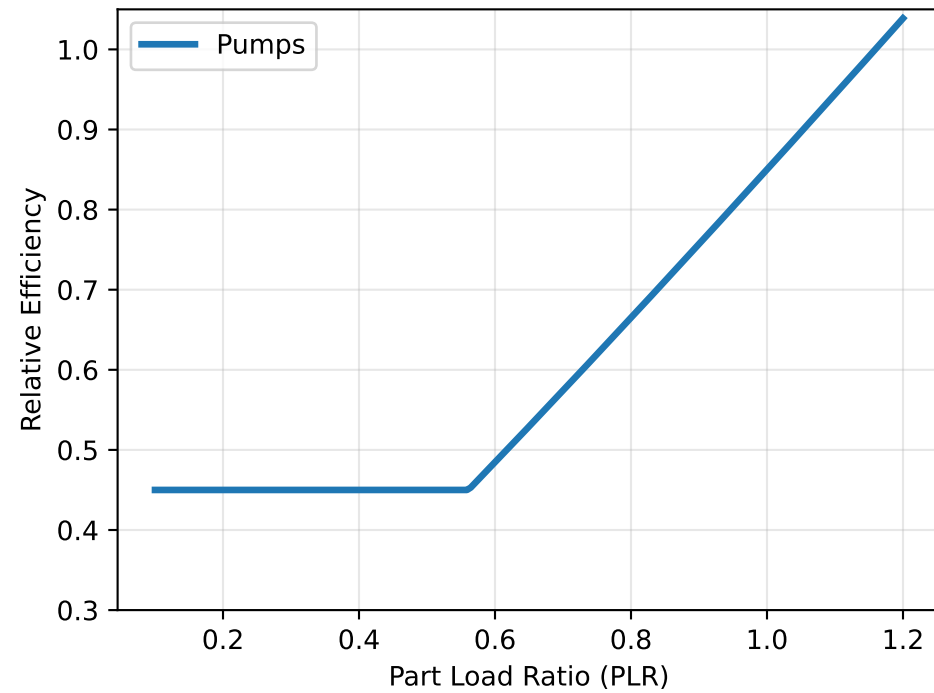
Part-Load Efficiency Curve: Boiler



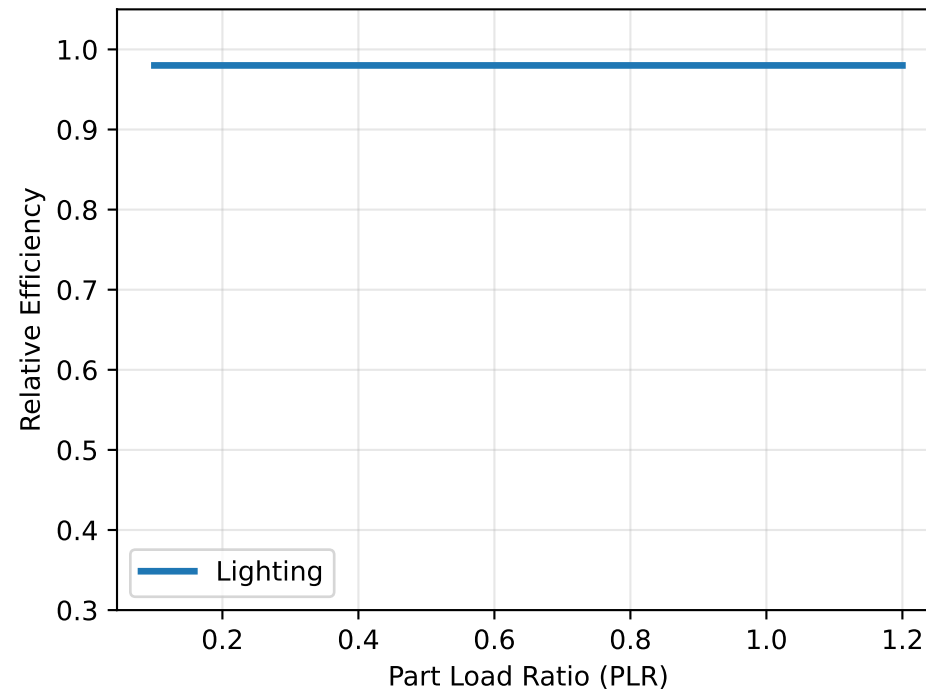
Part-Load Efficiency Curve: Fans



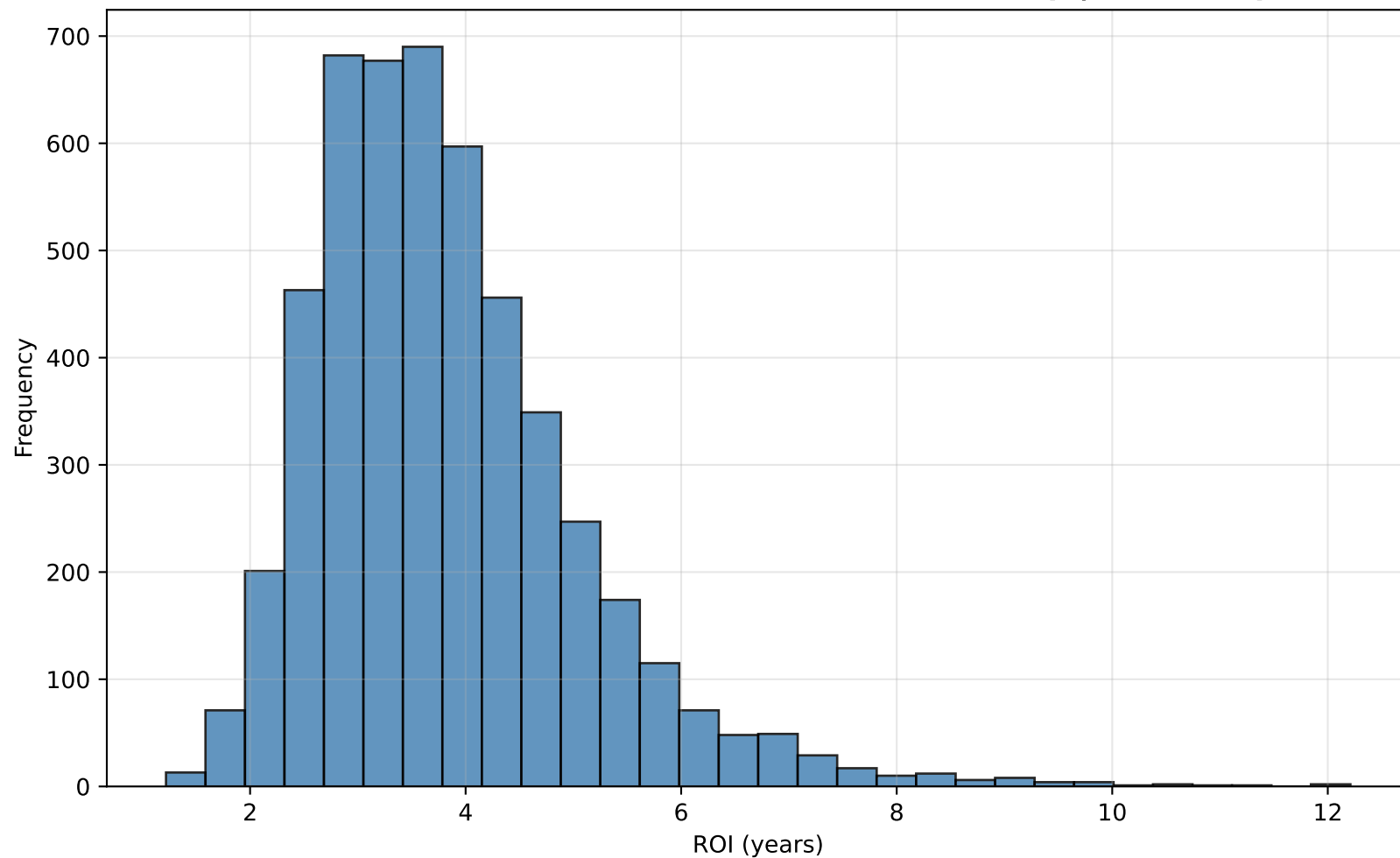
Part-Load Efficiency Curve: Pumps



Part-Load Efficiency Curve: Lighting



Monte Carlo Simulation — ROI Distribution (5,000 runs)



Optimal Configuration from Parametric Optimization

Recommended Upgraded Component Ratings:

Chiller	:	22.5 kW	(25.0% reduction)
Boiler	:	12.0 kW	(20.0% reduction)
Fans	:	3.2 kW	(36.0% reduction)
Pumps	:	3.5 kW	(30.0% reduction)
Lighting	:	3.2 kW	(60.0% reduction)